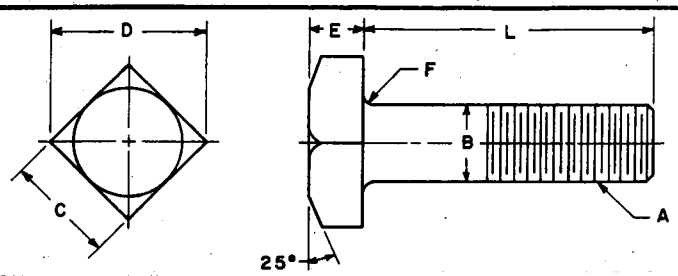
REVIEWER: WC, SH, 69, IS, NSA
USER: MO, GL, MC, WP, YD

This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

FED. SUP CLASS
Ⓐ5306



	NOMINAL SIZE		1/4	5/16	3/8	7/16	1/2
A	THREADS PER INCH		20UNC-2A	18UNC-2A	16UNC-2A	14UNC-2A	13UNC-2A
B	BODY DIAMETER	Max	.260	.324	.388	.452	.515
		Min	.2127	.2712	.3287	.3850	.4435
C	WIDTH ACROSS FLATS	Max	.3750	.5000	.5625	.6250	.7500
		Min	.362	.484	.544	.603	.725
D	WIDTH ACROSS CORNERS	Max	.530	.707	.795	.884	1.061
		Min	.498	.665	.747	.828	.995
E	HEAD HEIGHT	Max	.188	.220	.268	.316	.348
		Min	.156	.186	.232	.278	.308
F	RADIUS	Max	.031	.031	.031	.031	.031

L LENGTH	TOLERANCE 3/8 & UNDER	TOLERANCE 7/16 & 1/2	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN
3/4			1	010-9145	32		63	010-9180				
1			2	010-9147	33	010-9165	64	010-9182	94		124	010-9216
1-1/4			3	010-9149	34		65	010-9184	95		125	
1-1/2			4	010-9150	35	425-8542	66		96		126	010-9219
2		±1/16	5	010-9152	36	010-9170	67	010-9187	97		127	010-9221
2-1/2			6	010-9154	37		68	010-9189	98		128	010-9223
3			7	010-9156	38		69	010-9191	99	270-9787	129	010-9225
3-1/2	±1/32		8	010-9158	39		70		100		130	010-9227
4			9	010-9159	40	010-9177	71	010-9194	101		131	010-9228
4-1/2			10	010-9160	41		72	010-9195	102		132	010-9229
5			11	010-9161	42		73	010-9196	103		133	
5-1/2			12	010-9820	43	010-9821	74	270-9780	104		134	010-9824
6			13	010-9825	44		75	010-9827	105		135	010-9829
6-1/2			14		45		76	010-9832	106		136	261-2016
7			15	010-9835	46		77	010-9837	107		137	
7-1/2			16		47		78		108		138	
8			17	268-1494	48		79	270-9783	109		139	260-4491
8-1/2			18		49		80		110		140	
9			19		50		81	270-9784	111		141	
9-1/2			20		51		82		112		142	
10			21	276-8168	52		83	270-9786	113		143	425-0562
11		±3/32	22		53		84		114		144	
12			23		54		85	021-7813	115		145	270-9788
13	±1/16		24		55		86		116		146	
14			25		56		87		117		147	
15			26		57		88		118		148	
16			27		58		89		119		149	
17			28		59		90		120		150	
18			29		60		91		121		151	
19			30		61		92		122		152	
20			31		62		93		123		153	
21											154	
22											155	
23											156	
24											157	
25											158	

Bolts are (unfinished), not machined or equivalent on any surface except in the threads.
Bolts with shoulder next to the head are acceptable with manufacturers option.
There may be a reasonable swell or fin under the head not to exceed the nominal body diameter by the following: .030 for sizes thru 1/2", .050 for sizes 5/8 and 3/4", .063 for sizes over 3/4" thru 1-1/4".

MATERIAL: Steel, SAE Grade 1 or ASTM A-307, Grade A, 55,000 PSI minimum tensile strength.

PROTECTIVE COATING: Uncoated.

THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H-28.

THREAD LENGTH (MIN): (Extreme point to last complete thread) shall be twice the diameter plus 1/4 in. for lengths up to and including 6 in. and twice the diameter plus 1/2 in. for lengths over 6 in. Bolts too short for formula, thread lengths shall be threaded as close to the head as practicable.

NOTES: (1) Referenced documents of issue in effect on date of invitations for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example:MS35354-1.
(4) All dimensions in inches.

Ⓐ CANCELLED
SEE SHEET 2

Ⓐ For changes see sheets 1 and 2.

APPROVED 24 JANUARY 1955 REVISED Ⓐ 28 JULY 1966

P.A. WC	TITLE	MILITARY STANDARD
Other Cust SH 69	BOLT, MACHINE, SQUARE HEAD, (UNFINISHED), STEEL, UNCOATED, UNC-2A	MS 35354
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 2

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FED. SUP CLASS
Ⓐ 5306

	NOMINAL SIZE		5/8	3/4	7/8	1	1-1/8	1-1/4
A	THREADS PER INCH		11UNC-2A	10UNC-2A	9UNC-2A	8UNC-2A	7UNC-2A	7UNC-2A
B	BODY DIAMETER	Max	.642	.768	.895	1.022	1.149	1.277
		Min	.5589	.6773	.7946	.9100	1.0228	1.1476
C	WIDTH ACROSS FLATS	Max	.9375	1.1250	1.3125	1.5000	1.6875	1.8750
		Min	.906	1.088	1.269	1.450	1.631	1.812
D	WIDTH ACROSS CORNERS	Max	1.326	1.591	1.856	2.121	2.386	2.652
		Min	1.244	1.494	1.742	1.991	2.239	2.489
E	HEAD HEIGHT	Max	.444	.524	.620	.684	.780	.876
		Min	.400	.476	.568	.628	.720	.812
F	RADIUS	Max	.062	.062	.062	.093	.093	.093

L LENGTH	TOLERANCE	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN
1-1/2	±1/8	159		192		230		268					
2		160	010-1664	193	010-2484	231		269					
2-1/2		161	174-4151	194	010-2486	232		270					
3		162	010-1668	195		233		271		306		341	
3-1/2		163	010-1670	196	281-8306	234		272		307		342	
4		164	010-1671	197	010-2491	235		273		308		343	
4-1/2		165	281-8283	198	010-2492	236		274	021-7869	309		344	
5		166	010-1673	199		237		275	429-6576	310		345	
5-1/2		167	010-1674	200		238		276		311		346	
6		168		201	010-2495	239	430-9451	277		312		347	
6-1/2	±3/16	169		202	271-8169	240		278		313		348	
7		170	010-2475	203	010-2497	241		279		314		349	
7-1/2		171		204	011-9636	242		280		315		350	
8		172		205		243		281		316		351	
8-1/2		173		206		244		282		317		352	
9		174	011-9659	207		245		283	021-8049	318		353	
9-1/2		175	012-4721	208		246		284	270-9793	319		354	
10		176		209	174-4142	247		285		320		355	
11		177	270-9789	210		248		286		321		356	
12		178		211		249		287		322		357	
13		179		212		250		288		323		358	
14		180		213	426-6906	251		289		324		359	
15		181		214		252		290		325		360	
16		182		215		253		291		326		361	
17		183		216		254		292		327		362	
18		184		217	281-8285	255		293		328		363	
19		185		218		256		294		329		364	
20		186		219		257		295		330		365	
21		187		220		258		296		331		366	
22		188		221	270-9792	259		297		332		367	
23		189		222		260		298		333		368	
24		190		223		261		299		334		369	
25		191		224		262		300	270-9794	335		370	
26				225		263		301		336		371	
27				226		264		302		337		372	
28				227		265		303		338		373	
29				228		266		304		339		374	
30				229		267		305		340		375	

NOTE: For illustration, material, protective coating and other pertinent data see sheet 1.

Ⓐ CANCELLED AFTER 28 JULY 1966. USE MS35355. INTERCHANGEABILITY RELATIONSHIP - MS35355 can replace the cancelled bolts of the same size. The cancelled MS35354 cannot always replace MS35355 and should be used until existing stock is depleted. Then use only the superseding MS bolts of the same size for design and replacement.

Ⓐ CANCELLED
SEE NOTE

REVIEWER: WC, SH, 69, IS, NSA
USER: MO, GL, MC, WP, YD

APPROVED 24 JANUARY 1955 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC	TITLE	MILITARY STANDARD
Other Cust SH 69	BOLT, MACHINE, SQUARE HEAD, (UNFINISHED), STEEL, UNCOATED, UNC-2A	MS35354
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF

DD FORM SEP 67 672-1 (Coordinated)

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